



Lecture 2 – Linear Regression

Presented By

Md. Shoab Alam (Lecturer)

Green University of Bangladesh

Department of CSE

supervised learning

Input data



Annotations

These are
apples



Model

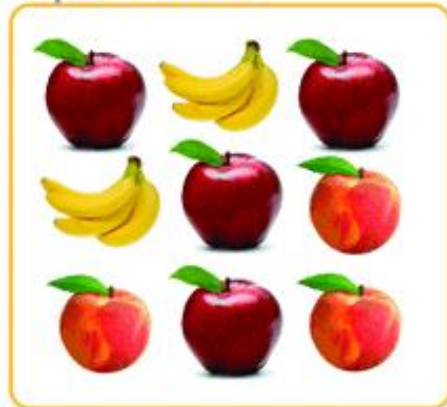


Prediction

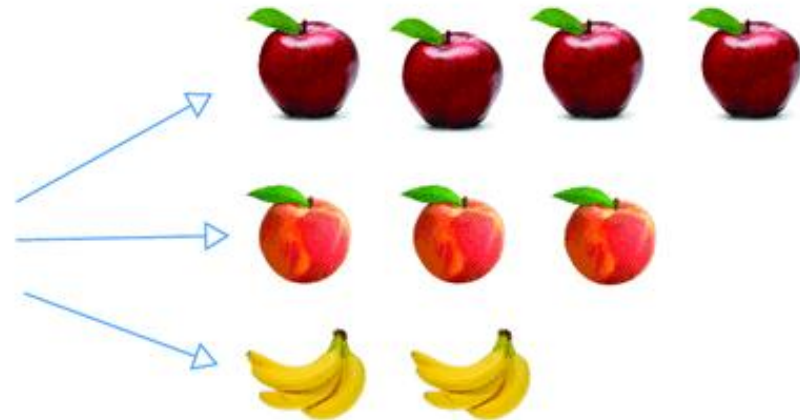


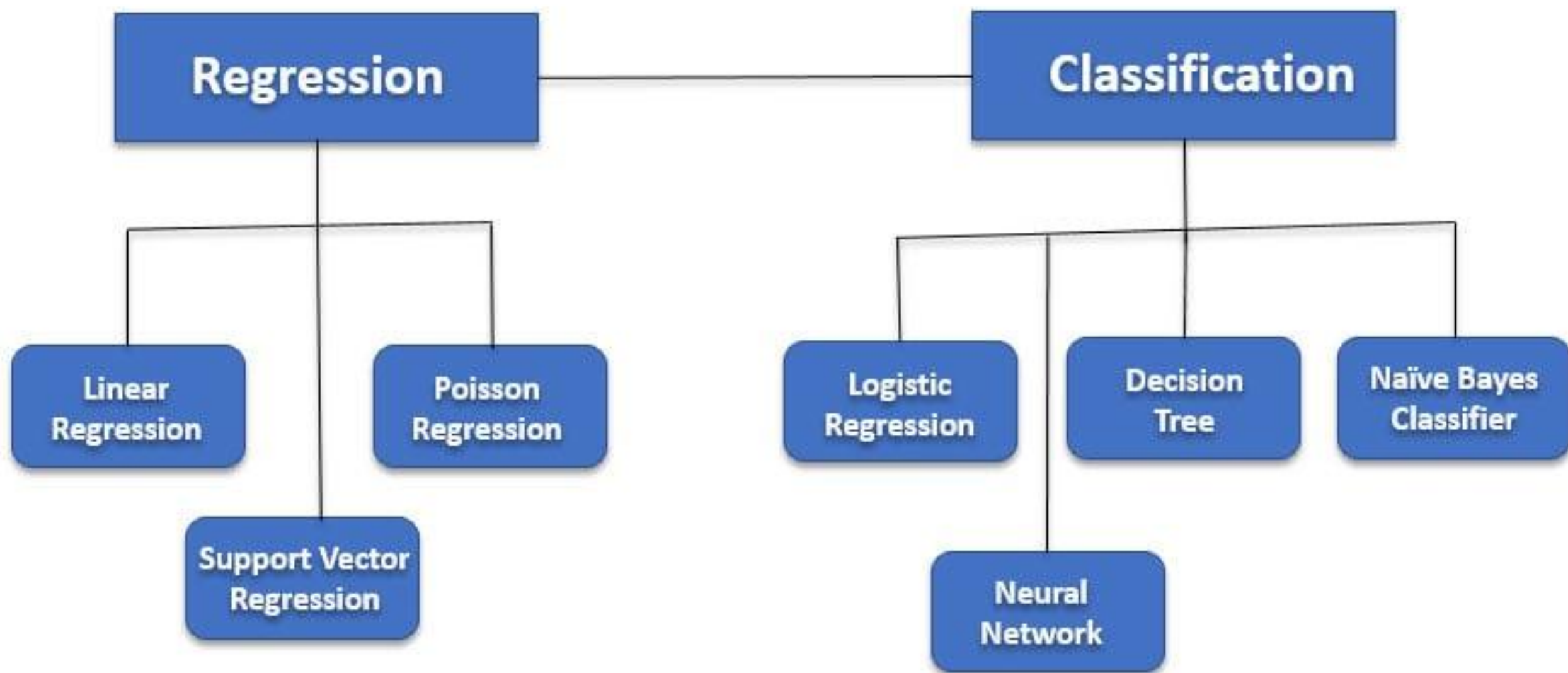
unsupervised learning

Input data



Model





Type of LR

Types of Linear Regression Models



Table 1-1. Does money make people happier?

Country	GDP per capita (USD)	Life satisfaction
Hungary	12,240	4.9
Korea	27,195	5.8
France	37,675	6.5
Australia	50,962	7.3
United States	55,805	7.2

Let's plot the data for these countries (Figure 1-17).

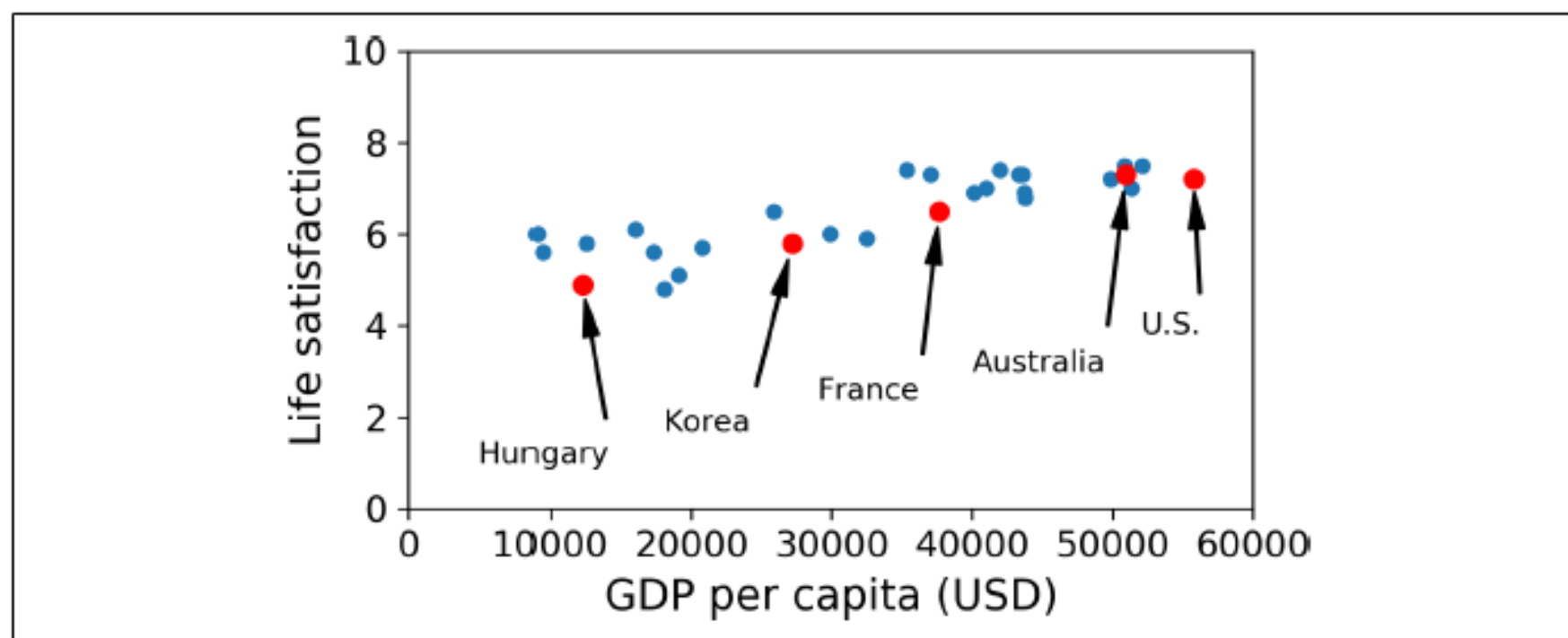
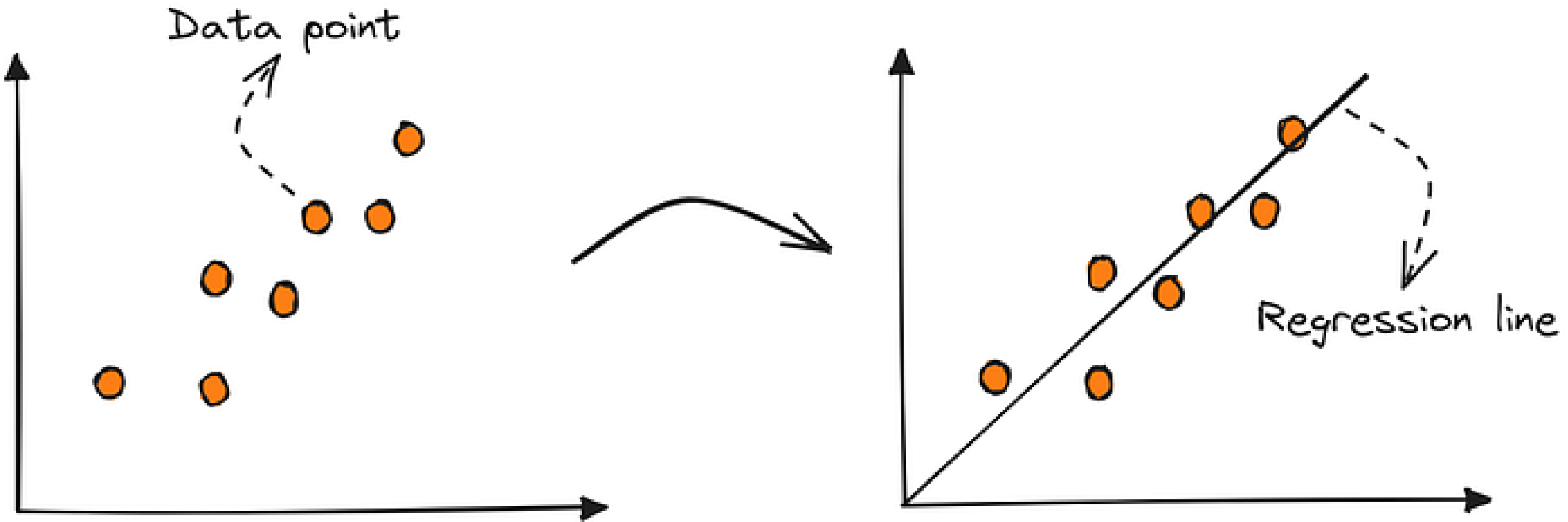
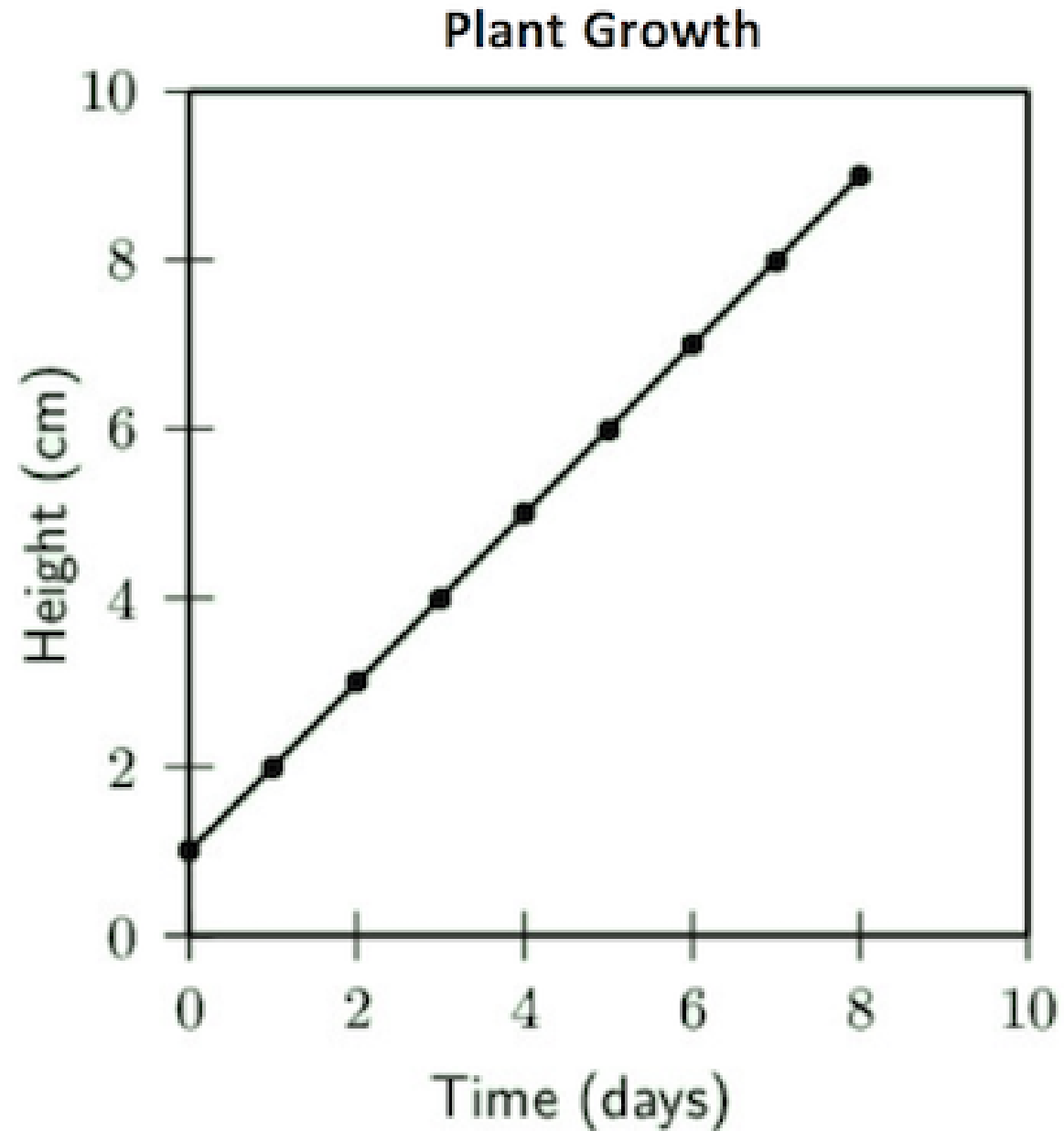


Figure 1-17. Do you see a trend here?

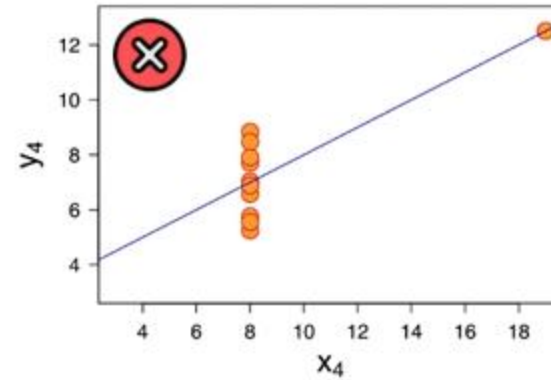
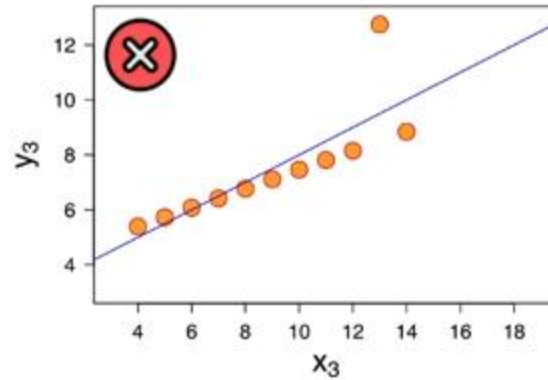
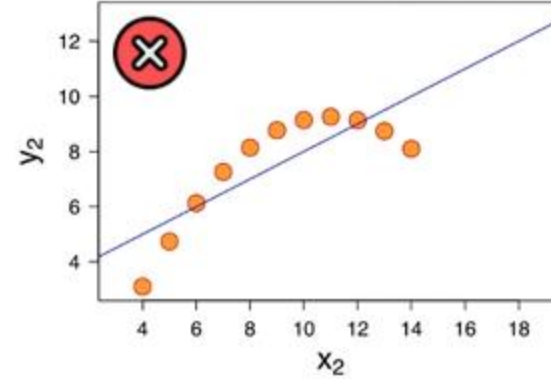
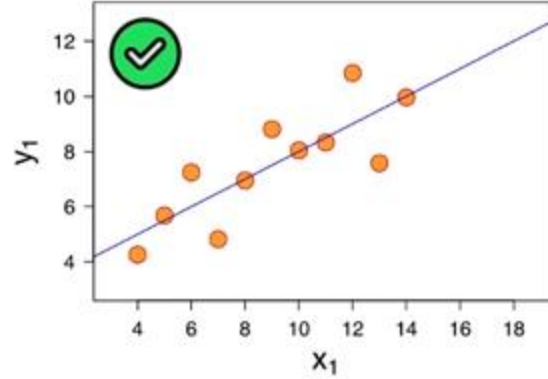
Sort of Linear Data



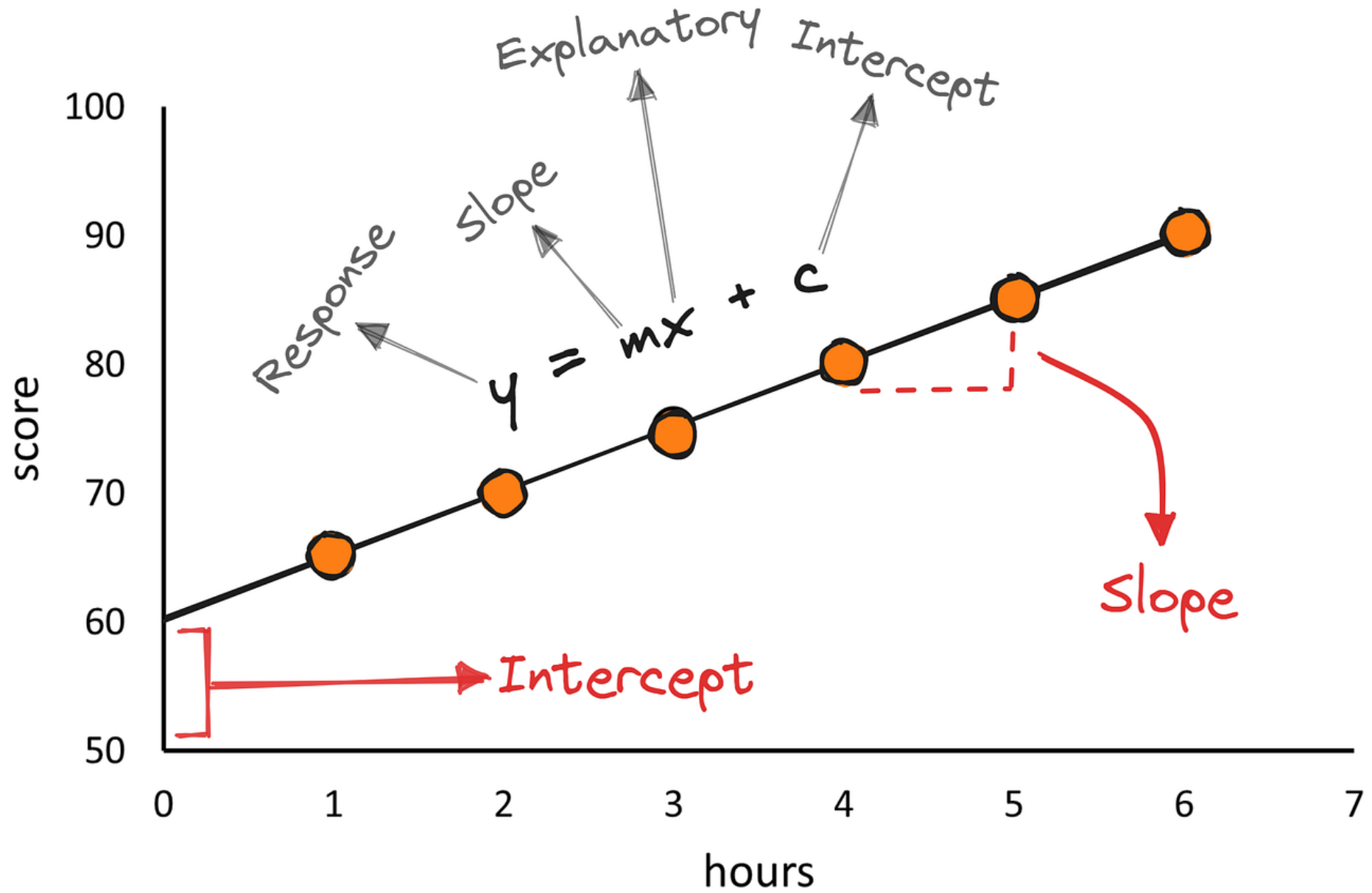
Linear Data



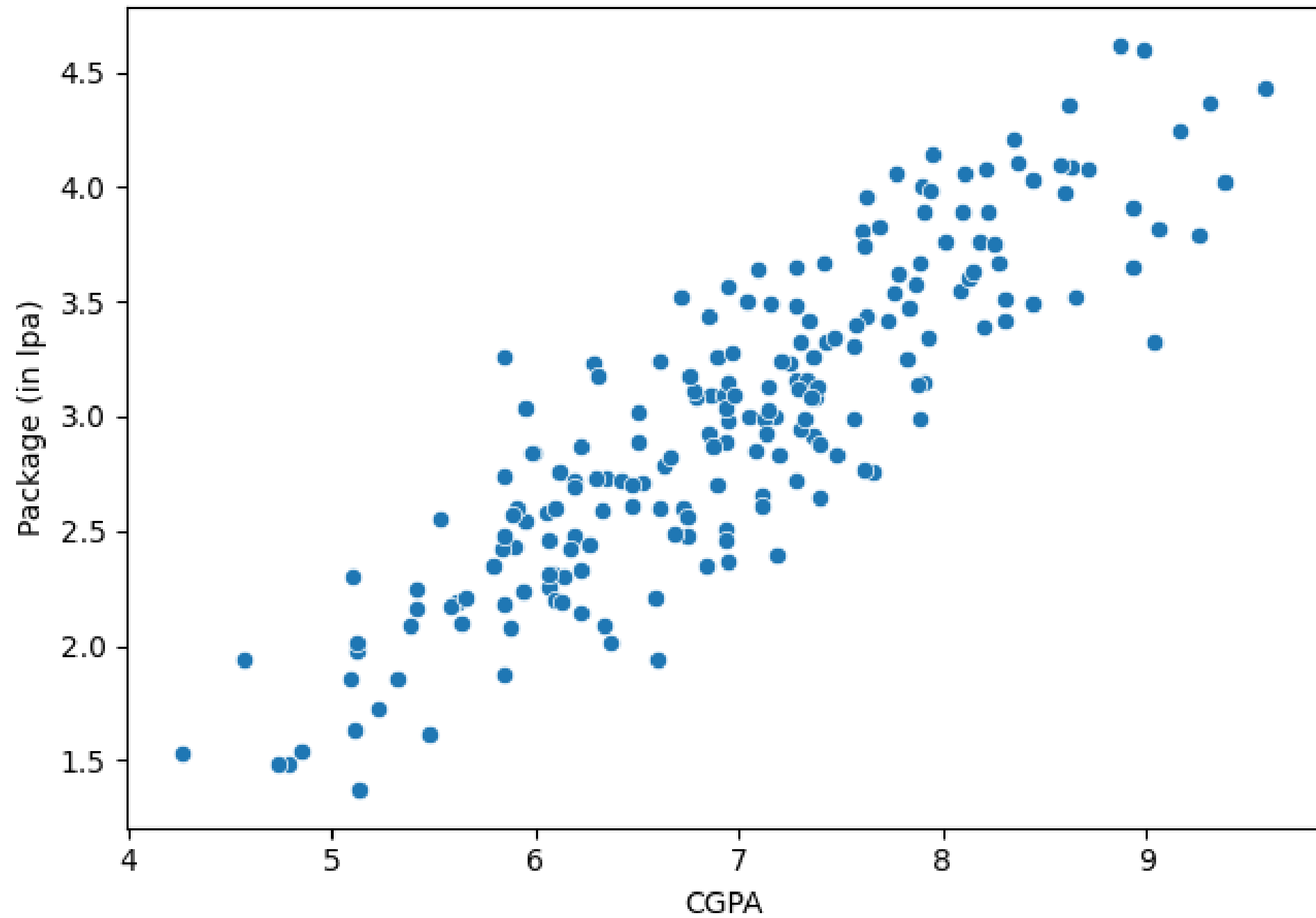
Assumptions of Linear Regression



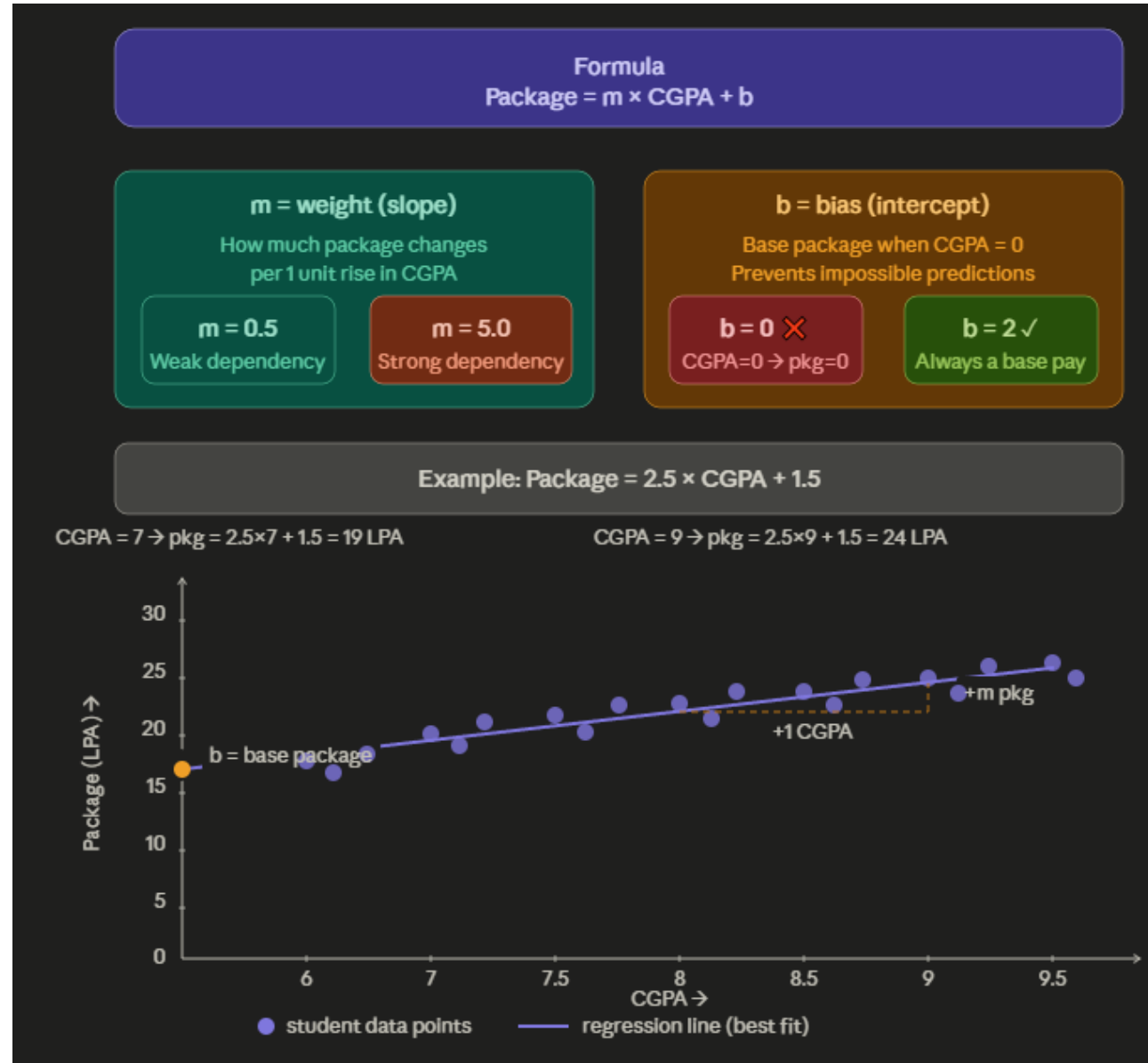
Best Fit Line



CGPA vs Package



Linear Regression Parameters



Linear Regression Parameters

- © m is the weight (slope). It tells you how sensitive the package is to CGPA. If $m = 2.5$, every 1-point rise in CGPA adds 2.5 LPA to the package. A small m (like 0.2) means CGPA barely moves the needle — maybe experience matters more. A large m (like 5) means CGPA is a strong predictor — the model leans heavily on it.
- © b is the **bias** (intercept). It is the package the model predicts even before CGPA contributes anything. This is crucial because without it, $\text{Package} = m \times \theta = 0$ when CGPA is zero — which is nonsensical.
- © **Intuition test:**
 - m answers: *"how much does this feature matter?"*
 - b answers: *"what's the floor, independent of the feature?"*

Mathematics

- To Calculate the best m, b we have two different solutions –
 1. Closed Form Solution – Direct Formula (Ordinary Least Squares)
 2. Non-Closed Form Solution - Gradient Descent for higher dimensional data.

✓ **Formula for intercept (b):**

$$b = \bar{y} - m\bar{x}$$

Where:

- $\bar{x}$ is the mean of x
- $\bar{y}$ is the mean of y
- n is the number of data points

✓ **Formula for slope (m):**

$$m = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

Alternatively:

$$m = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$

Linear Regression



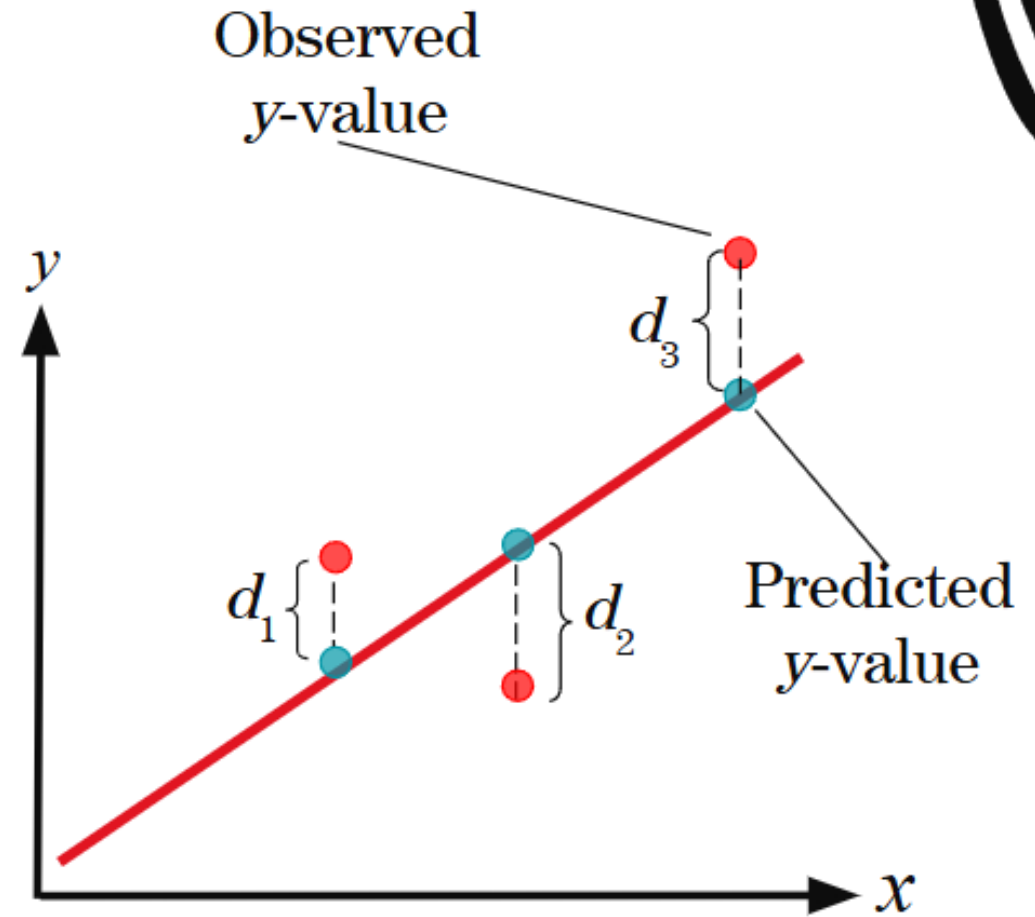
Real = Y



Predicted = $\hat{Y}$

$$\text{Error } d_1 = Y_i - \hat{Y}$$

$$\text{Total Error} = \sum (Y_i - \hat{Y})^2$$



Linear Regression

$$\text{Total Error} = \sum (Y_i - (mx + b))^2$$

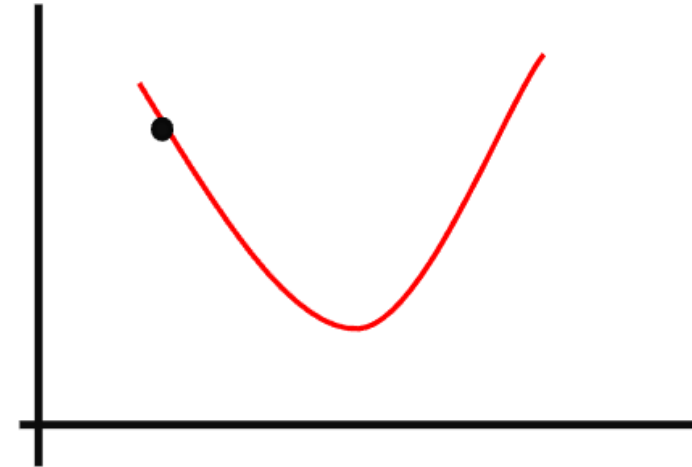
$$y = mx + b$$

Sum of Square Error

$$E(m, b) = \sum_{i=1}^n (Y_i - (mx_i - b))^2$$

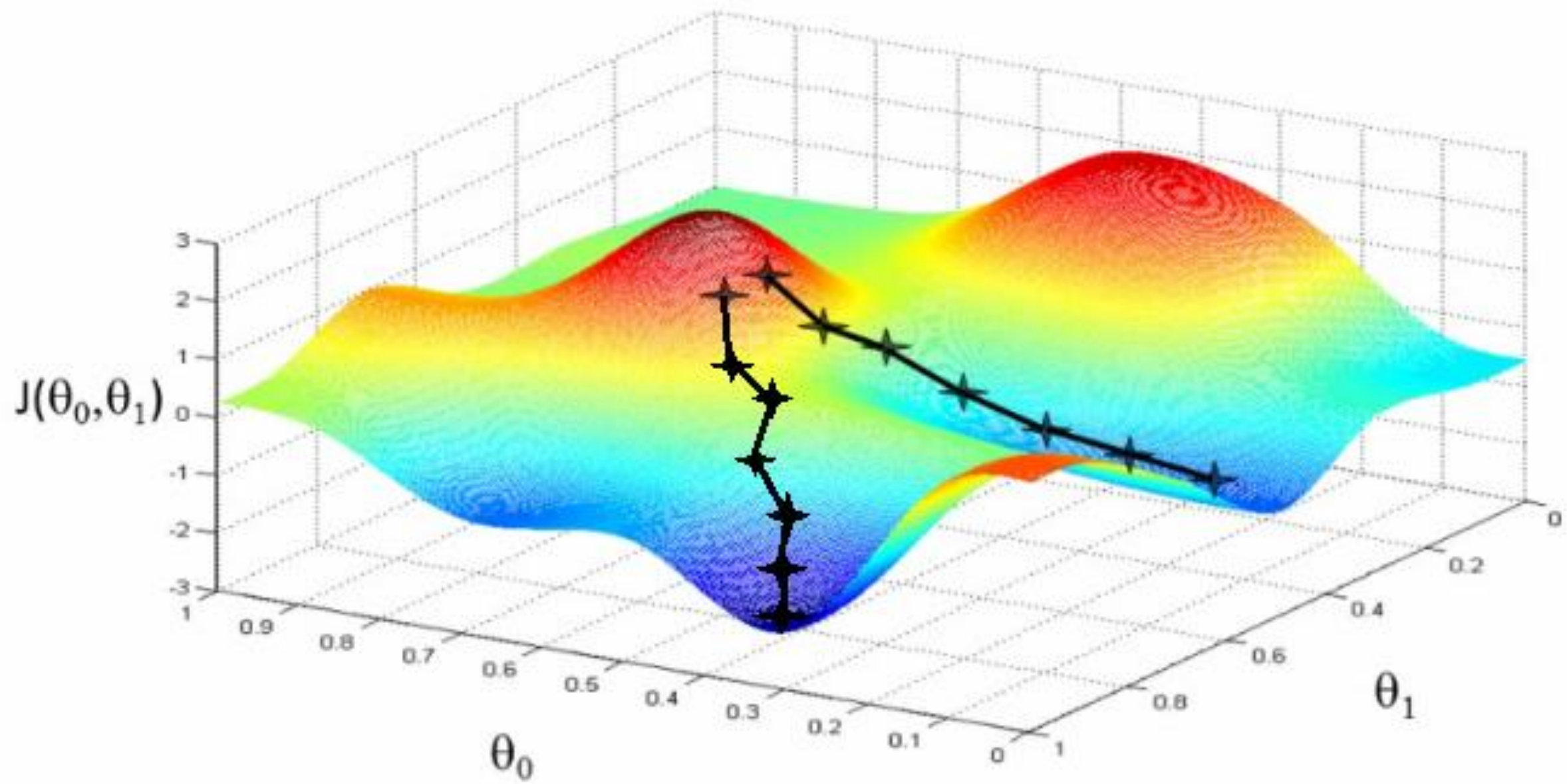
$$E(m) = \sum (Y_i - mx_i)^2, \text{ assuming } b=0$$

$$E(b) = \sum (Y_i - x_i - b)^2, \text{ assuming } m=1$$



(i) Partial Derivative with Respect to b

$$\frac{\partial E}{\partial b} = \frac{\partial}{\partial b} \sum_{i=1}^n (y_i - (b + mx_i))^2$$



Linear Regression

(i) Partial Derivative with Respect to $b \rightarrow$ Deriving $b = \bar{y} - m \bar{x}$

Start:
$$\frac{\partial E}{\partial b} = \partial / \partial b \sum (y_i - (b + mx_i))^2$$

Given SSE

Step 1:
$$\frac{\partial E}{\partial b} = -2 \sum (y_i - b - mx_i)$$

Chain rule on $(y-b-mx)^2$ gives $-2(y-b-mx)$

Step 2:
$$-2 \sum (y_i - b - mx_i) = 0$$

Set derivative = 0 to minimize

Step 3:
$$\sum (y_i - b - mx_i) = 0$$

Divide both sides by -2

Step 4:
$$\sum y_i - nb - m \sum x_i = 0$$

$\sum b = nb$ (b is a constant)

Step 5:
$$(1/n) \sum y_i - b - m \cdot (1/n) \sum x_i = 0$$

Divide entire equation by n

Step 6:
$$\bar{y} - b - m \bar{x} = 0 \implies \mathbf{b = \bar{y} - m \bar{x}}$$

$\bar{y} = \sum y_i / n$, $\bar{x} = \sum x_i / n$

Linear Regression

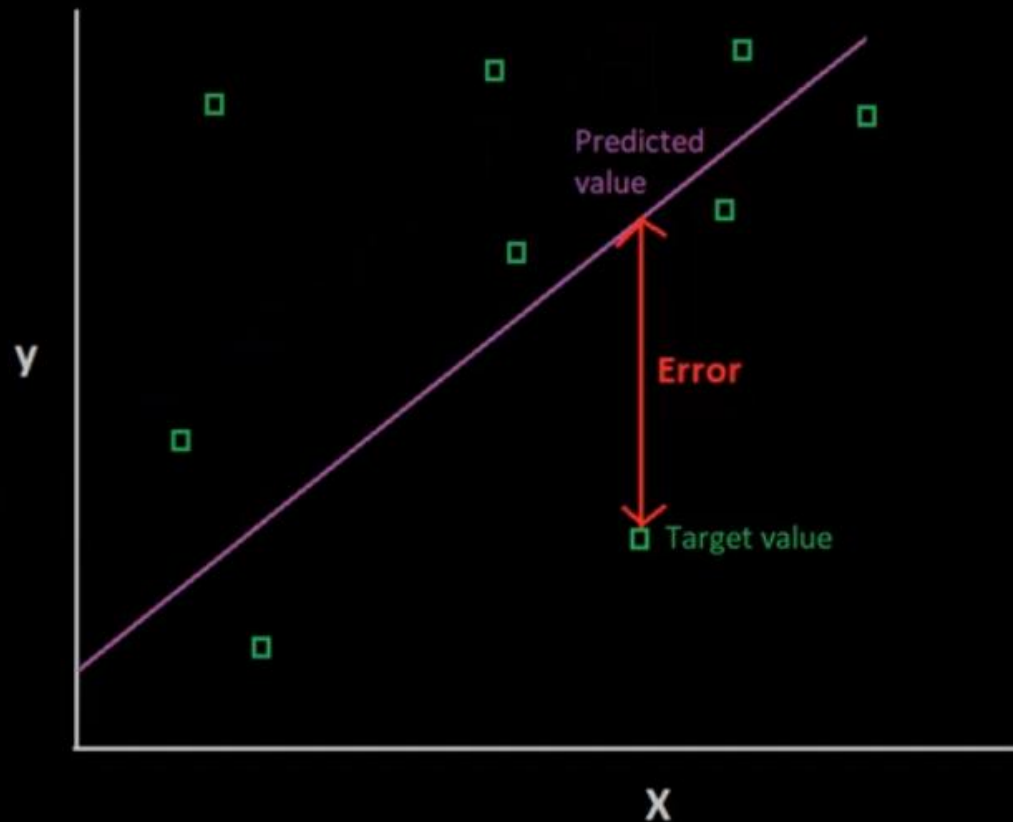
(ii) Partial Derivative with Respect to $m \rightarrow$ Substitute $b = \bar{y} - m\bar{x}$ first, then differentiate

Start:	$E = \sum (y_i - mx_i - \bar{y} + m\bar{x})^2$	<i>Sub $b = \bar{y} - m\bar{x}$ into $E = \sum (y_i - b - mx_i)^2$</i>
Step 1:	$\frac{\partial E}{\partial m} = \sum 2(y_i - mx_i - \bar{y} + m\bar{x})(-x_i + \bar{x}) = 0$	<i>Chain rule: d/dm of $(\dots)^2 = 2(\dots) \cdot d/dm(\dots)$</i>
Step 2:	$\Rightarrow -2 \sum (y_i - mx_i - \bar{y} + m\bar{x})(x_i - \bar{x}) = 0$	<i>$(-x_i + \bar{x}) = -(x_i - \bar{x})$; factor out -2</i>
Step 3:	$\Rightarrow \sum (y_i - mx_i - \bar{y} + m\bar{x})(x_i - \bar{x}) = 0$	<i>Divide both sides by -2</i>
Step 4:	$= \sum [(y_i - \bar{y}) - m(x_i - \bar{x})](x_i - \bar{x}) = 0$	<i>Regroup: $y_i - mx_i - \bar{y} + m\bar{x} = (y_i - \bar{y}) - m(x_i - \bar{x})$</i>
Step 5:	$= \sum [(y_i - \bar{y})(x_i - \bar{x}) - m(x_i - \bar{x})^2] = 0$	<i>Distribute $(x_i - \bar{x})$ into bracket</i>
Step 6:	$\sum (y_i - \bar{y})(x_i - \bar{x}) = m \sum (x_i - \bar{x})^2$	<i>Separate Σ and move m term to RHS</i>

Result:	$m = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$
----------------	--

Mean Square Error (MSE)

$$\text{MSE} = \frac{1}{N} \sum \left(\text{Target value} - \text{Predicted Value} \right)^2$$



Evaluation Metrics

- Root Mean Square Error
- Mean Absolute Error
- R^2

Root Mean Square Error

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y - y_{pred})^2}$$

RMSE tells us how far apart the predicted values are from the observed values in a dataset, on average. The lower the RMSE, the better a model fits a dataset

Mean Absolute Error

$$MAE = \frac{1}{n} \sum_{i=1}^n |y - y_{pred}|$$

R^2

$$R^2 = 1 - \frac{SSE}{SST} = 1 - \frac{\sum_{i=1}^n (y - y_{pred})^2}{\sum_{i=1}^n (y^* - y_{mean})^2}$$

This value ranges from 0 to 1.

The higher the R^2 value, the better a model fits a dataset.

R^2 will be negative if the model prediction is bad i.e the sum of square error is large as compare to sum of total error.

Multiple Linear Regression



$$\hat{y} = b_0 + b_1X_1 + b_2X_2 + \cdots + b_nX_n$$

Dependent variable

y-intercept
(constant)

Slope coefficient 1

Independent variable 1

Slope coefficient 2

Independent variable 2

Slope coefficient n

Independent variable n



